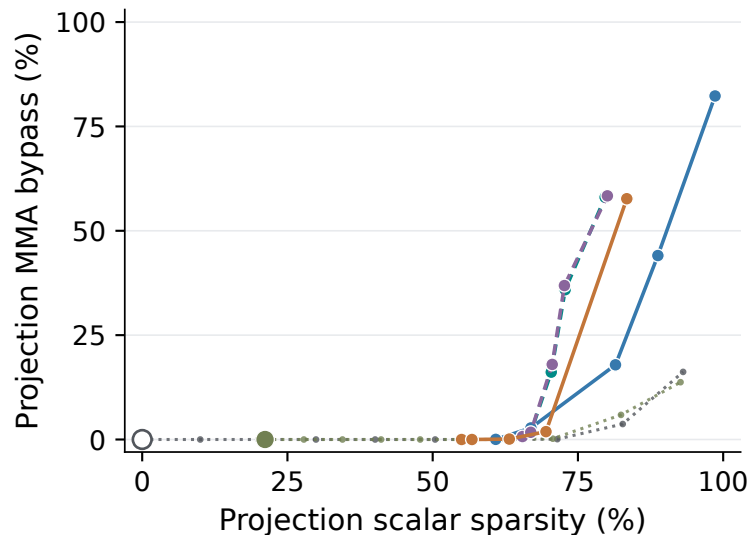
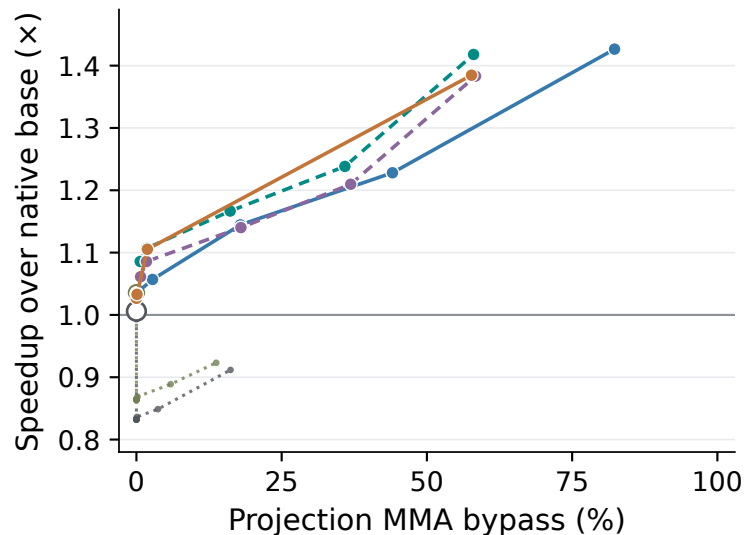


Scalar sparsity, instruction bypass and base-model speedup

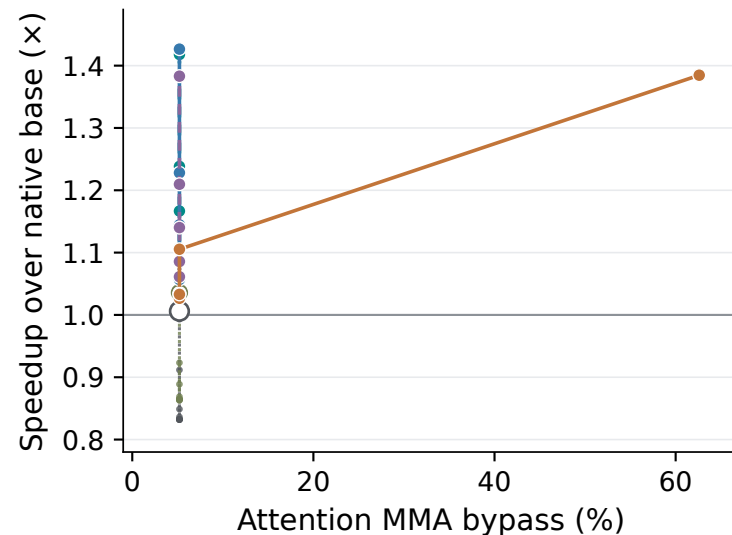
(a) 14M: Sparsity and bypass



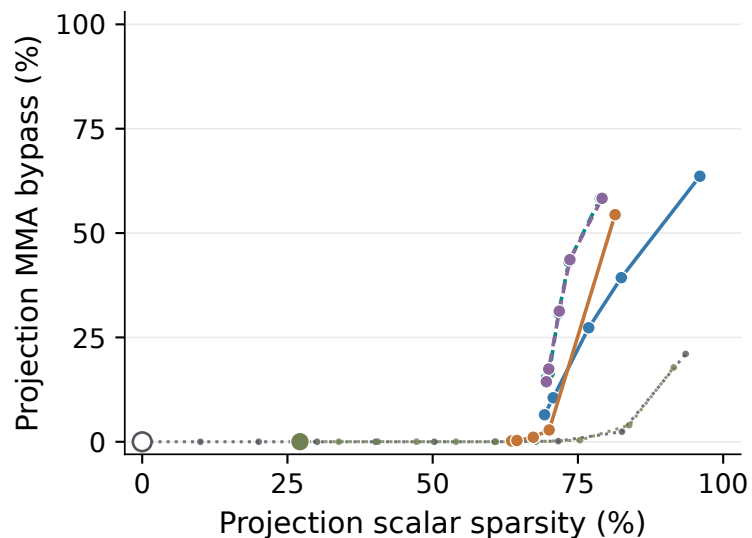
(b) 14M: Projection bypass and speedup



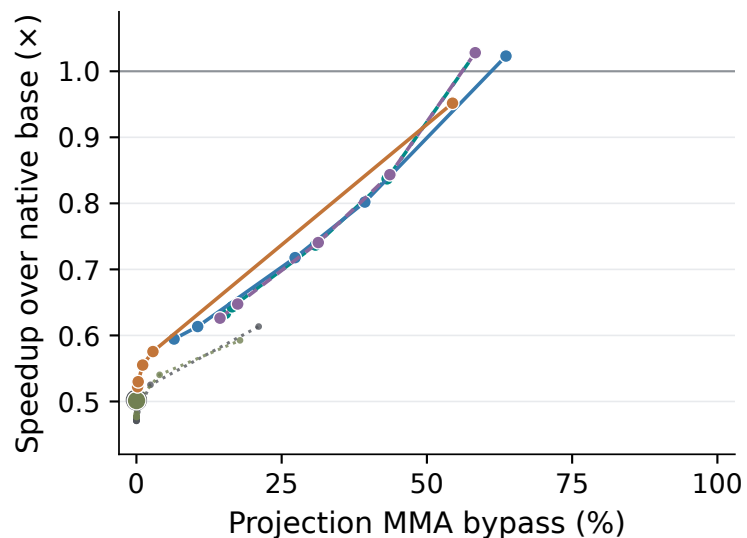
(c) 14M: Attention bypass and speedup



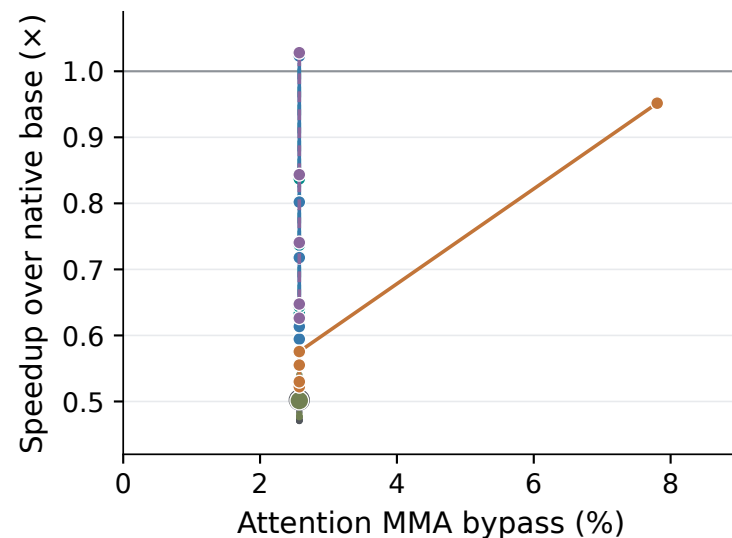
(d) 70M: Sparsity and bypass



(e) 70M: Projection bypass and speedup



(f) 70M: Attention bypass and speedup



- Base model
- GeLU → ReLU
- T_4/P_h
- T_7/P_h
- T_4/P_{all}
- T_7/P_{all}

Dotted paths: post-hoc clipping of the base and ReLU models